

A Deterministic Critical Besov Lower Bound for White Noise When $1 \leq p < 2$

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Status and source question

This is a `candidate_full_likely_valid` solution to Remark 3.5(i) of Mark Veraar, *Regularity of Gaussian white noise on the d -dimensional torus*, arXiv:1010.6219 [1]. Theorem 3.4(2) of the source proves that for every $p \in [2, \infty)$ there is $c_{p,d} > 0$ such that

$$\mathbb{P}\left(\|W\|_{B_{p,\infty}^{-d/2}(\mathbb{T}^d)} \geq c_{p,d}\right) = 1.$$

Remark 3.5(i), on PDF page 7, asks whether this remains true for $p \in [1, 2)$ and notes that Hausdorff–Young is insufficient.

(2) For all $p \in [2, \infty)$ there exists a constant $c_{p,d}$ such that

$$\mathbb{P}(\|W\|_{B_{p,\infty}^{-d/2}(\mathbb{T}^d)} \geq c_{p,d}) = 1.$$

(3) For any $p, q \in [1, \infty]$ and $s < -d/2$ one has $\mathbb{P}(W \in B_{p,q}^s(\mathbb{T}^d)) = 1$.

(4) For any $p \in [1, \infty]$ and $q \in [1, \infty)$ one has $\mathbb{P}(W \in B_{p,q}^{-d/2}(\mathbb{T}^d)) = 0$.

(5) For any $p, q \in [1, \infty]$ and $s > -d/2$ one has $\mathbb{P}(W \in B_{p,q}^s(\mathbb{T}^d)) = 0$.

(6) One has $\mathbb{P}(W \in B_{\infty,\infty}^{-d/2}(\mathbb{T}^d)) = 0$.

Remark 3.5. Some remarks on the theorem:

(i) We do not know whether (2) holds for $p \in [1, 2)$ as well. Applying the Hausdorff–Young’s inequality instead of Parseval’s identity is not sufficient here.

(ii) The proof shows that one can take $c_{2,d} = \sqrt{2^{3d/2} - 2^{-3d/2}}$ in (2).

Proof of Theorem 3.4. Before we present the proof we reduce to a particular choice of the Gaussian white noise. Let $W = \sum_{k \in \mathbb{Z}^d} \gamma_k e_k$ in $\mathcal{D}'(\mathbb{T}^d)$. Then by Proposition 3.3, W is a Gaussian white noise. Let $V : \Omega \rightarrow \mathcal{D}'(\mathbb{T}^d)$ be an arbitrary white noise, and let $\mathcal{W}$ and $\mathcal{V}$ be as in Lemma 3.2. Fix $p, q \in [1, \infty]$. As in (2.3) let

$$(3.1) \quad W_j(x) = \langle \varphi_{j,x}, W \rangle = \mathcal{W}(\varphi_{j,x}) \text{ and } V_j(x) = \langle \varphi_{j,x}, V \rangle = \mathcal{V}(\varphi_{j,x}) \quad j \in \mathbb{N}.$$

Now one has

Definitions and idea of the proof

Use the source’s Littlewood–Paley function ϕ , supported in $\{\frac{1}{2} \leq |\xi| \leq 2\}$ and equal to one when $2^{-1/2} \leq |\xi| \leq 2^{1/2}$. Put $\varphi_j(k) = \phi(2^{-j}k)$. For the canonical white noise

$$W = \sum_{k \in \mathbb{Z}^d} \gamma_k e^{ik \cdot x}, \quad W_j(x) = \sum_{k \in \mathbb{Z}^d} \varphi_j(k) \gamma_k e^{ik \cdot x},$$

where the γ_k are independent standard complex Gaussians, the critical norm is

$$\|W\|_{B_{p,\infty}^{-d/2}} = \sup_{j \geq 0} 2^{-jd/2} \|W_j\|_{L^p(\mathbb{T}^d)}.$$

The source obtains its lower bound for $p \geq 2$ by comparing L^p with L^2 and using Parseval. That comparison reverses below two. Instead, regard the normalized L^p mass itself as a nonnegative random variable. Its first moment stays uniformly away from zero and its second moment stays uniformly bounded. Paley–Zygmund therefore gives a scale-independent positive chance of a fixed lower bound. Every third dyadic block uses disjoint Gaussian coefficients, so Borel–Cantelli upgrades that chance to an almost-sure infinitely-often statement.

Main result

Theorem 1 (low- p lower bound). *Let $d \geq 1$, let W be a Gaussian white noise on $\mathbb{T}^d$, and use the Littlewood–Paley decomposition fixed above. For every $p \in [1, \infty)$ there exists a deterministic constant $c_{p,d} > 0$ such that*

$$\mathbb{P}\left(\|W\|_{B_{p,\infty}^{-d/2}(\mathbb{T}^d)} \geq c_{p,d}\right) = 1.$$

In particular, the answer to Remark 3.5(i) is affirmative for every $p \in [1, 2)$.

Proof. By Lemma 3.2 and the reduction at the beginning of the proof of Theorem 3.4 in [1], the Besov norm of an arbitrary white noise has the same law as that of the canonical series above. It is therefore enough to prove the claim for that series.

Fix $p \in [1, \infty)$ and set

$$Z_j = 2^{-jd/2} W_j, \quad Y_j = \|Z_j\|_{L^p(\mathbb{T}^d)}^p.$$

For every $x \in \mathbb{T}^d$, unitary invariance of the standard complex Gaussian shows that $Z_j(x)$ has the same law as $v_j^{1/2} \gamma$, where γ is a standard complex Gaussian and

$$v_j = 2^{-jd} \sum_{k \in \mathbb{Z}^d} |\varphi_j(k)|^2.$$

This variance is independent of x . The annulus

$$S_j = \{k \in \mathbb{Z}^d : 2^{j-1/2} < |k| \leq 2^{j+1/2}\}$$

lies where $\varphi_j = 1$, while φ_j is supported in $\{k : 2^{j-1} \leq |k| \leq 2^{j+1}\}$. Elementary lattice counting therefore gives constants $a_d, b_d > 0$ and j_0 such that

$$0 < a_d \leq v_j \leq b_d < \infty \quad (j \geq j_0). \quad (1)$$

Let $V_d = |\mathbb{T}^d| = (2\pi)^d$ and $m_r = \mathbb{E}|\gamma|^r$. Tonelli’s theorem and Gaussian scaling give

$$\mathbb{E}Y_j = \int_{\mathbb{T}^d} \mathbb{E}|Z_j(x)|^p dx = V_d m_p v_j^{p/2} \geq V_d m_p a_d^{p/2} =: A_{p,d} > 0. \quad (2)$$

For the second moment, Cauchy–Schwarz in probability gives, for all x, y ,

$$\mathbb{E}(|Z_j(x)|^p |Z_j(y)|^p) \leq (\mathbb{E}|Z_j(x)|^{2p} \mathbb{E}|Z_j(y)|^{2p})^{1/2} = m_{2p} v_j^p.$$

After integration and (1),

$$\mathbb{E}Y_j^2 \leq V_d^2 m_{2p} b_d^p =: B_{p,d} < \infty. \quad (3)$$

No independence between the values at x and y has been used.

Paley–Zygmund, (2), and (3) imply

$$\begin{aligned}\mathbb{P}\left(Y_j \geq \frac{A_{p,d}}{2}\right) &\geq \mathbb{P}\left(Y_j \geq \frac{1}{2}\mathbb{E}Y_j\right) \\ &\geq \frac{(\mathbb{E}Y_j)^2}{4\mathbb{E}Y_j^2} \geq \frac{A_{p,d}^2}{4B_{p,d}} =: \delta_{p,d} > 0 \quad (j \geq j_0).\end{aligned}$$

It remains to turn this uniform probability into probability one. The Fourier supports of φ_{3n} are pairwise disjoint: the support for $3n$ is contained in $[2^{3n-1}, 2^{3n+1}]$ in radial frequency, whereas the next begins at 2^{3n+2} . Hence the random fields $(Z_{3n})_n$, and therefore the events

$$E_n = \{Y_{3n} \geq A_{p,d}/2\},$$

are independent. Since $\mathbb{P}(E_n) \geq \delta_{p,d}$ for all large n , the second Borel–Cantelli lemma gives $\mathbb{P}(E_n \text{ infinitely often}) = 1$. On that event,

$$\|W\|_{B_{p,\infty}^{-d/2}} = \sup_{j \geq 0} \|Z_j\|_p \geq \left(\frac{A_{p,d}}{2}\right)^{1/p}.$$

Thus one may take $c_{p,d} = (A_{p,d}/2)^{1/p} > 0$. Equality in distribution transfers the conclusion to every Gaussian white noise. $\square$

Verification, scope, and novelty status

The proof needs no numerical experiment. The accompanying verification report separately audits the pointwise Gaussian law, the two moment bounds, the direction of Paley–Zygmund, disjointness of the $3n$ supports, and the equidistribution reduction. In particular, the endpoint $p = 1$ causes no loss: all positive Gaussian moments are finite.

The constant displayed here is not claimed sharp. It depends on the fixed cutoff through the lower lattice-count constant; equivalence of admissible Besov norms transfers the qualitative deterministic lower bound to any other standard cutoff.

Eight focused attempts are recorded in the attempt note. The decisive upgrade was to replace the failed deterministic L^2 comparison by a two-moment small-ball argument, then recover almost sure behavior on an independent shell subsequence.

Cheap run indexes and bounded web/arXiv searches on 17 August 2026 used the exact Remark 3.5(i) sentence, the paper title and author, and the phrases *low p critical Besov white noise lower bound* and *Paley–Zygmund dyadic white noise Besov*. They found the source and its published version but no explicit later resolution. Novelty confidence is moderate-high, not definitive.

Human-review recommendation. Verify the multiplier-support convention for the independent $3n$ subsequence and perform a broader specialist literature search. The moment and Borel–Cantelli argument is otherwise self-contained.

References

- [1] M. Veraar, *Regularity of Gaussian white noise on the d-dimensional torus*, Banach Center Publ. 95 (2011), 385–398; arXiv:1010.6219.